Lukas Trottner

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Employment

Junior Professor (TT), head of the Mathematical Statistics group, **University of Stuttgart** since Oct. 2025
Assistant Professor, School of Mathematics, **University of Birmingham** Sep. 2024 – Oct. 2025
Postdoc, Department of Mathematics, **Aarhus University** Sep. 2021 – Sep. 2024

Education

University of Mannheim

PhD in Mathematics (Dr. rer. nat.) 2018 – 2021
Thesis: *Implications of Markov stability theory for nonparametric statistics, Markov additive fluctuations and data-driven stochastic control*, Advisor: Prof. Dr. Leif Döring, grade: summa cum laude

MSc in Mathematics in Business and Economics 2016 – 2018
Thesis: *Lamperti transformation for self-similar Markov processes*, grade: 1.0

BSc in Mathematics in Business and Economics 2012 – 2016
Thesis: *Verzweigungsprozesse: Aussterben einer Population*, grade: 1.1

Research Interests

Generative models, statistics for stochastic processes, concentration of measure, transfer learning, MCMC, data-driven stochastic optimal control, high-dimensional statistical inference, Lévy processes and Markov additive processes

Publications

Journal articles

Statistical guarantees for denoising reflected diffusion models (with Asbjørn Holk and Claudia Strauch)
Journal of Machine Learning Research 27 (2026), paper no. 116, pp. 1–50. URL: http://jmlr.org/papers/v27/25-1588.html

Change point estimation for a stochastic heat equation (with Markus Reiß and Claudia Strauch)
Annals of Statistics 54 (2026), no. 1, pp. 277–299, DOI: 10.1214/25-AOS2567

Beyond Fixed Horizons: A Theoretical Framework for Adaptive Denoising Diffusions
(with Sören Christensen, Jan Kallsen and Claudia Strauch)
Transctions on Machine Learning Research (2026), URL: https://openreview.net/forum?id=xowTpMf9fM

Multivariate change estimation for a stochastic heat equation from local measurements (with Anton Tiepner)
Stochastic Processes and their Applications 192 (2026), paper no. 104832, DOI: 10.1016/j.spa.2025.104832

Mini-Workshop: Statisticial Challenges for Deep Generative Models
(with Sören Christensen, Alain Oliviero-Durmus and Claudia Strauch)
Oberwolfach Reports 22 (2025), no. 1, pp. 375–403, DOI: 10.4171/owr/2025/8

Data-driven rules for multidimensional reflection problems (with Sören Christensen and Asbjørn Holk Thomsen)
SIAM/ASA Journal on Uncertainty Quantification 12 (2024), no. 4, pp. 1240–1272, DOI: 10.1137/23M1618570

Markov additive friendships (with Leif Döring and Alex Watson)
Transactions of the American Mathematical Society 377 (2024), no. 11, pp. 7699–7752, DOI: 10.1090/tran/9266

The uniqueness of the Wiener–Hopf factorisation of Lévy processes and random walks
(with Leif Döring, Mladen Savov and Alex Watson)
Bulletin of the London Mathematical Society 56 (2024), no. 9, pp. 2951–2968, DOI: 10.1112/blms.13112

Covariate shift in nonparametric regression with Markovian design
Information and Inference 13 (2024), no. 2, paper no. iaae011. DOI: 10.1093/imaiai/iaae011

Learning to reflect: A unifying approach for data-driven stochastic control strategies (with S. Christensen and C. Strauch)
Bernoulli 30 (2024), no. 3, pp. 2074–2101. DOI: 10.3150/23-BEJ1665

Stability of overshoots of Markov additive processes (with Leif Döring)
Annals of Applied Probability 33 (2023), no. 6B, pp. 5413–5458. DOI: 10.1214/23-AAP1951

Concentration analysis of multivariate elliptic diffusions (with Cathrine Aeckerle-Willems and Claudia Strauch)
Journal of Machine Learning Research 24 (2023), paper no. 106, pp. 1–38. URL: http://jmlr.org/papers/v24/22-0666.html.

Adaptive invariant density estimation for continuous-time mixing Markov processes under sup-norm risk
(with Niklas Dexheimer and Claudia Strauch)
Annales de l'Institut Henri Poincaré B (Probab. et Stat.) 58 (2022), no. 4, pp. 2029–2064. DOI: 10.1214/21-aihp1235

Conference papers

Statistical Convergence of Spherical First Hitting Diffusion Models (with Simon Bienewald)
Advances in Neural Information Processing Systems (NeurIPS 2026) (accepted)

Interdisciplinary work

F. V. Jensen, J. P. Carvalho, N. Wili, A. Holk Thomsen, D. L. Goodwin, L. Trottner, C. Strauch, A. Bodholt Nielsen and N. Chr. Nielsen.
In situ Learning-Based Spin Engineering of Pulsed Dynamic Nuclear Polarization.
Science Advances 12 (2026), no. 38, pp. eaeh4467. DOI: 10.1126/sciadv.aeh4467

Preprints

Reflected diffusion models adapt to low-dimensional data (2026)
with Asbjørn Holk and Claudia Strauch, arXiv:2603.24495

Model-free filtering in high dimensions via projection and score-based diffusions (2025)
with Sören Christensen, Jan Kallsen and Claudia Strauch, arXiv:2510.23197

Awards and Scholarships

Award for best master's degree in Mathematics, University of Mannheim	2018
Award for best master's thesis in Mathematics, University of Mannheim	2018
Award for best bachelor's degree in Mathematics, University of Mannheim	2016
Deutschlandstipendium, scholarship, University of Mannheim	2012 – 2018

Teaching

University of Stuttgart

Lecturing

Generative Models	Winter 2026
Mathematische Statistik	Summer 2026
Stochastik und angewandte Mathematik für das Lehramt (joint with Benjamin Stamm)	Winter 2025

Supervision

Bachelor thesis on Fundamentals of probabilistic language models	Summer 2026

University of Birmingham

Lecturing

Bayesian Inference and Computation (joint with Mengchu Li)	Spring 2025
Foundations of Statistical Inference (joint with Mengchu Li)	Fall 2024

Supervision

8 MSc projects in Financial Risk Management	Summer 2025

Aarhus University

Lecturing

Statistics for Stochastic Processes	Spring 2024
Statistics for Stochastic Processes	Spring 2022

Tutoring

Introduction to Probability Theory and Statistics	Fall 2023
Introduction to Probability Theory and Statistics	Fall 2022
Calculus	Fall 2021

University of Mannheim

Supervision

Master thesis on Q-Learning and stochastic approximation	Spring 2021

Tutoring

Introduction to Statistical Learning Theory	Fall 2019
Analysis II	Spring 2019
Introduction to Mathematical Statistics	Fall 2018
Advanced Mathematical Finance	Spring 2018
Mathematical Finance	Fall 2017

Supervision of PhD students

Simon Bienewald (University of Stuttgart)	since Nov. 2025
Asbjørn Holk Thomsen (Aarhus University, joint with Claudia Strauch)	2023–2026

Thesis: *Reflected Diffusions and their Applications: From Data-driven Control to Generative Models*

Research Groups

Member of Research Training Group 1953 "Statistical Modeling of Complex Systems"	2018 – 2021
Research Assistant, Digital Economy, Leibniz Centre for European Economic Research	2012 – 2017

Professional Activities

Presentations and Seminars (past 5 years)

5th Italian Meeting on Probability and Mathematical Statistics (IMPMS), University of Palermo, *invited*	June 2026
16th Workshop on Stochastic Models, Statistics and Their Applications (SMSA), University of Würzburg, *invited*	March 2026
Oberseminar Probability Theory and Mathematical Statistics, University of Bielefeld, *invited*	Jan. 2026
IMS International Conference on Statistics and Data Science, Sevilla, *invited*	Dec. 2025
Probab. Persp. in Neural Network Based Machine Learn., Math. Forschungsinst. Oberwolfach, *invited*	Oct. 2025
11th International Conference on Lévy Processes, Sofia, *invited*	July 2025
Stochastic Partial Differential Equations: Statistics Meets Numerics, Institut Mittag-Leffler, *invited*	June 2025
Frontiers of Statistics and Machine Learning, Mathematisches Forschungsinstitut Oberwolfach, *invited*	March 2025
SNU Workshop: Nonparametric Inference in the Age of AI, Seoul National University, *invited*	Feb 2025
Statistical Challenges for Deep Generative Models, Math. Forschungsinst. Oberwolfach, *invited*	Feb. 2025
Algorithms & Computationally Intensive Inference seminars, University of Warwick, *invited*	Nov. 2024
Statistics Seminar, University of Cambridge, *invited*	Nov. 2024
Mathematics Department Seminar, University of Siegen, *invited*	Nov. 2024
Data Science and Computational Statistics Seminar, University of Birmingham, *invited*	Nov. 2024
12th World Congress of the Bachelier Finance Society, Rio de Janeiro	July 2024
Institute of Statistics and Operations Research Colloquium, University of Vienna, *invited*	April 2024
Workshop on Stochastic Analysis and Statistics, Tokyo, *invited*	Feb. 2024
NeurIPS 2023, New Orleans, *poster*	Dec. 2023
43rd Conference on Stochastic Processes and their Applications, Lisbon, *two presentations*	July 2023
European Meeting of Statisticians 2023, Warsaw	July 2023
Informs APS 2023, Nancy, *invited*	June 2023
Stochastics Seminar, Aarhus University, *invited*	April 2023
German Probability and Statistics Days 2023, Essen	March 2023
SIAM Conference on Computational Science and Engineering 2023, Amsterdam, *invited*	March 2023
Seminar Department of Mathematics, Kiel University, *invited*	Nov. 2022
Workshop on Statistics for Stochastic Processes, Luxembourg, *invited*	Sep. 2022

10th International Conference on Lévy processes, Mannheim, *invited*	July 2022
International Symposium on Nonparametric Statistics, Paphos, *invited*	June 2022
Stochastics Seminar, Aarhus University, *invited*	Nov. 2021
German Probability and Statistics Days 2021, Mannheim	Sep. 2021
IRTG "Stochastic Analysis in Interaction", Berlin Mathematical School, *invited*	June 2021

Invited lecture series

Foundations of SPDEs at the workshop "Pathways into Mathematics of SPDEs", Heidelberg University	March 2026

Organization of scientific events

MFO Mini-Workshop on "Statistical Challenges for Deep Generative Models", Oberwolfach	Feb. 2025
Dynstoch Satellite Summer School, Kiel	May 2024
Contributed session "High-dimensional statistics for stochastic processes" at SPA 2023, Lisbon	July 2023
The Midsummer Lectures 2023	June 2023
Aarhus–Copenhagen Data Science Meeting 2022	Dec. 2022

Research visits

Heidelberg University, host: Claudia Strauch	Nov. 2024
University College London, host: Alex Watson	March 2023
Kiel University, host: Sören Christensen	Nov. 2022
Humboldt-Universität zu Berlin, host: Markus Reiß	Oct. 2022
Humboldt-Universität zu Berlin, host: Markus Reiß	Aug. 2022

Summer/Spring/Fall Schools

SFB1294 Spring School on Data Assimilation, Döllnsee	March 2022
RTG1953 Fall School on Causal Inference and Stein's method, Mannheim	Nov. 2019
Satellite Summer School to the 9th International Conference on Lévy Processes, Athens	July 2019

Refereeing

Annals of Statistics, Journal of Machine Learning Research, Bernoulli, NeurIPS, Information and Inference, Annales de l'Institut Henri Poincaré (Probab. et Stat.), Stochastic Processes and their Applications, Electronic Journal of Statistics, Electronic Journal of Probability, Electronic Communications in Probability, Biometrika, Inverse Problems, ALEA, Advances/Journal of Applied Probability, Annales Henri Lebesgue, Annals of the Institute of Statistical Mathematics, Statistics

Membership in Learned Societies

Deutsche Mathematikervereinigung

Vita Date: 25 September 2026.